

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS:

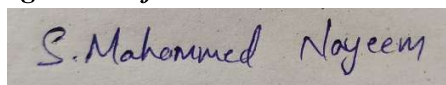
Total Marks: 20

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I S.Mahammed Nayee /192425294 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

(i) Machine Learning vs Deep Learning:-

Machine Learning:- Raw data

↓
Data preprocessing

↓
Manual Feature engineering

↓
Machine learning algorithm

↓
Prediction / output.

Deep Learning:-

Raw data

↓
Input layer

↓
Hidden layer

↓
Output layer.

Advantages:-

* Eliminates manual feature engineering.

* Learns hierarchical representations.

* Improves performance on complex datasets.

* Handles high-dimensional.

Real-world applications:-

Machine learning:-

1. Email spam detection - Gmail filter spam emails.
2. Credit card fraud detection - Banks detect fraudulent.

Deep learning:-

1. Face recognition - Smartphone face unlock systems.
2. Self-driving cars - Detect roads, pedestrians and traffic signs.

Comparison between ML and DL

Aspect	Machine learning	Deep learning
Architecture	Traditional algorithms	Multi-layer neural networks
Data requirement	Small to medium	Large datasets
Feature extraction	Manual	Automatic
Training Time	Faster	Slower
Hardware	CPU	GPU/TPU
Accuracy	Good	Very High

(2)

The concept of representation learning:-

Introduction:-

Representation learning is a technique in which neural networks automatically discover useful features from raw data instead of relying on manually engineered features.

Representation learning Architecture:-

Raw Image



Input layer



Hidden layers



Output layer

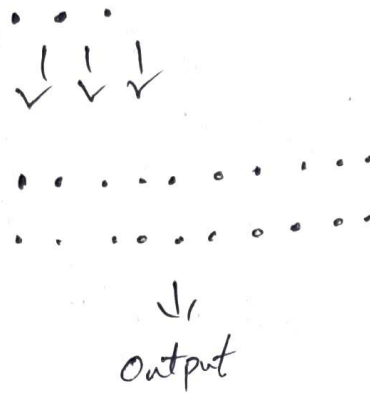
Advantages of Representation learning:-

- * Eliminates manual feature engineering.
- * Learns hierarchical features.
- * Improves accuracy.
- * Handles high dimensional data
- * Generalizes well to unseen data.

width of neural network!

width refers to the number of neurons in each hidden layer.

Architecture:



Effects:

- * learns more features simultaneously.
- * Higher memory consumption.
- * Better approximation of simple functions.
- * May cause overfitting if too wide.

Depth of Neural network!

Depth of refers to the number of hidden layers

Input

↓

layer 1

↓

layer 2

↓

layer 3

③

Compare

ReLU, Leaky

ReLU (LReLU)

and

ELU

Activation Functions

Introduction

Activation functions introduce non-linearity into neural networks, allowing them to learn complex relationships.

Architecture

Input



weighted sum



Activation Function

(ReLU / LReLU / ELU)



Output

1. ReLU (Rectified Linear unit)

Formula:

$$f(x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Mathematical Behaviour

* Outputs the input directly if it is positive.

* Returns zero for negative inputs.

Applications:-

- * Image classification.
- * Object detection using CNNs.

2. Leaky ReLU (LReLU):-

Formula:-

$$f(x) = \begin{cases} x, & x \geq 0 \\ 0.01x, & x < 0 \end{cases}$$

Mathematical Behaviour:-

- * Allows a small non-zero gradient for negative inputs.

Applications:-

- * GAN
- * Deep CNNs.

3. ELU (Exponential Linear unit)

Formula:-

$$f(x) = \begin{cases} x, & x \geq 0 \\ \alpha(e^x - 1), & x < 0 \end{cases}$$

Mathematical Behaviour:-

- * Positive values is passed unchanged.
- * Neurons receives weighted input.

Applications:-

- * Image classification
- * CNNs
- * Object detection.
- * Medical Image analysis
- * Face recognition.

(4)

Concept of unsupervised learning in neural networks

Introduction

- * Artificial intelligence (AI) enables computers to perform tasks that normally require humans.
- * One of the important branches of AI is ML.
- * Where computers learn patterns from data.

Architecture of Unsupervised learning

Unlabeled Data



Neural network model



Hidden feature learning



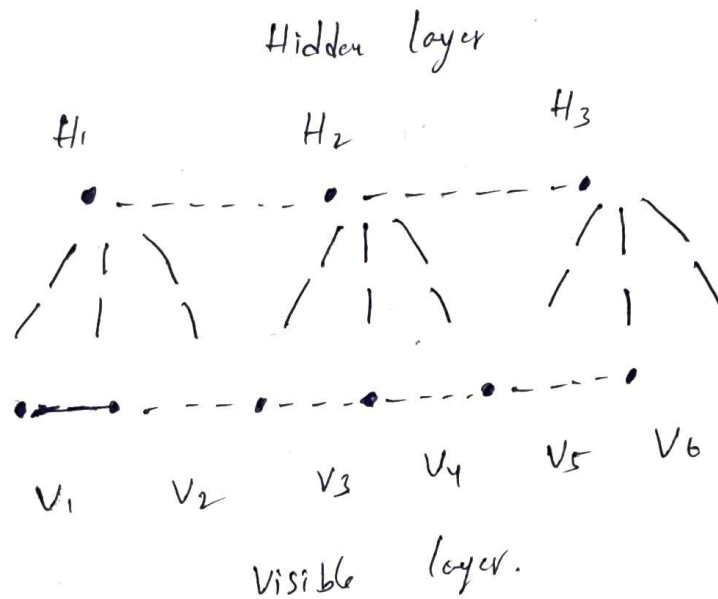
Pattern Discovery / clustering.

- * Uses only input data without target labels.
- * Learns hidden structures automatically.
- * Identifies similarities among data samples.
- * Discovers patterns without human supervision.
- * Performs automatic feature extraction.

Characteristics of unsupervised learning:-

- * Does not require labeled training data.
- * Learns meaningful representations automatically.
- * Identifies hidden relationships.
- * Performs clustering and feature learning.
- * Handles high dimensional datasets efficiently.

Architecture of RBM:-



Conclusion:-

- * Unsupervised learning enables neural networks to learn from unlabeled data.
- * Restricted Boltzmann Machines (RBMs) are probabilistic models mainly used for feature learning.